Fiscal Year:	FY 2024	**Task Last Updated:**	FY 09/02/2025
PI Name:	Cheng, Qingsu Ph.D.		
Project Title:	Evaluate Breast Cancer Risks During Long-Term Space-Travel by Using Mammographic Density Model		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53211-3029	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	UNSOL-JSC
Start Date:	07/01/2023	**End Date:**	06/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2025 per J. Weeks/SR (Ed., 4/2/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC23K0989		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	During a round trip to Mars, astronauts will be exposed to a significant amount of radiation, which can increase their cancer risk. Female astronauts are more susceptible to radiation because the breast tissue is radiosensitive. Our initiative improves the risk model of breast cancer for women, during a long-term spaceflight, with high mammographic density (MD), which is the second most important risk factor after BRCA1/2 mutations. High MD is rendered by the increased stiffness of the microenvironment. Our preliminary results show that in vitro models of high MD have increased frequencies of aberrant colony organization and DNA damage. But these frequencies increase with radiation exposure. Our validated assays are sensitive and quantitative to complement the statistical needs of radiation biology. In summary, our goal is to leverage MD as a risk factor to evaluate the tissue-specific risks of space radiation exposure on cancer outcomes and identify early surrogate biomarkers that correlate with cancer, pre-malignancy, or the hallmarks of cancer.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposed study will improve the understanding of breast cancer risk for female astronauts with high mammographic density (MD) for long-term stay in space.
Task Progress:	1.1. Published paper on IEEE Access. Shrestha A, Bao X, Cheng Q, McRoy S. CNN-Modified Encoders in U-Net for Nuclei Segmentation and Quantification of Fluorescent Images. IEEE Access. 2024. 12.10.1109/ACCESS.2024.3436509. This paper presents a deep learning approach for automated nuclei segmentation and quantification in fluorescent microscopy images by enhancing the U-Net architecture with powerful convolutional neural network encoders such as EfficientNet-B5/B7, ResNet-50, and Inception-ResNet-v2. The pipeline integrates post-processing steps to refine segmentation and perform detailed morphological quantification of each nucleus. This CNN-modified U-Net achieved an F1-score of 87% and an IoU of 80%, demonstrating both high accuracy and robust generalization to diverse, unseen datasets. The method offers a reliable and efficient tool for large-scale, automated analysis of fluorescent images, with applications in cell biology, pathology, and other biomedical research domains where precise nuclear measurements are essential. 1.2. Submitted paper to Scientific Data Fang J, Shrestha A, Zhao J, Cheng Q, A Comprehensive Dataset for Nuclei Instance Segmentation in DAPI-Stained Fluorescent Mammary Gland Images. Submitted to Scientific Data. The paper introduces the Breast Mammary Gland Dataset (BMGD), an annotated collection of 819 DAPI-stained image patches containing over 10,000 manually segmented nuclei from 3D organoid cultures exposed to various environmental stressors. Each image includes precise ground truth masks, providing a valuable resource for developing and evaluating automated nuclei segmentation algorithms, especially for breast cancer studies involving radiation exposure. 1.3. Manuscript in preparation (a) Our findings indicated a heightened susceptibility to radiation-induced morphological abnormalities. Typical colony formations after receiving daily radiation were shown in the columns of (Low Stiffness + 9 cGy) and (High Stiffness + 9 cGy) in Figure 1. High mammographic density (MD) has an impact on the disorganization of colony formation. The disorganized colony formation does not look different between radiation conditions, but the frequency changes. SPE radiation after shielding does not change much in the colony disorganization. (b) Morphological changes under radiation exposure: We have analyzed 184D's morphological changes using consensus clustering, and the results are shown in Figure 2. The results characterized a few morphological variables, such as Colony Elongation, Empty Volume Per Cell, Number of Cells Per Colony, and Physical Size Per Colony were changed toward abnormality under radiation. Colonies were more elongated, and bigger colonies were more likely formed in the radiation environment, while the lumen of each colony was shrinking under the exposure to radiation. We conclude that the chances of aberrant colony formation, or dysplasia, were increased by the radiation exposure. We have determined that for 184D, high mammographic density indeed is a risk factor for continuous exposure to low-dose radiation. There are individual differences in terms of radiation sensitivity; however, with the 3 lines we used, the difference is not significant in our current research setting. (c) DNA damages were observed that daily exposure kept short term DNA repair machinery continue to work. There are a lot of variations at the single cell level, indicating sensitivity at single cell level. Results were shown in Figure 3. Overall, there is evidence of individualized sensitivity among primary cells, though commercially available HMEC primary cells are limited. Induced pluripotent cells may provide future opportunities to study radiation sensitivity. Current findings do not indicate a clear link to cancer. However, transcriptomic analysis suggests potential for countermeasures, and AI could aid in identifying relevant molecules. Additionally, a regression-based prediction tool could be developed to connect colony disorganization with cell line genomes.
Bibliography Type:	Description: (Last Updated: 09/11/2025)
Abstracts for Journals and Proceedings	Jinwei Fang, Ashique Rahman, Zeyun Yu, and Qingsu Cheng. "From supervised learning to semi-supervised learning for improved efficiency of nuclei segmentation." 2025 Department of Energy (DOE) Biological Systems Science Division (BSSD) Meeting, Crystal City, Virginia, March 31-April 2, 2025. Abstracts. 2025 Department of Energy (DOE) Biological Systems Science Division (BSSD) Meeting, Crystal City, Virginia, March 31-April 2, 2025. , Apr-2025
Abstracts for Journals and Proceedings	Shrestha A, Bao X, Cheng Q, Mcroy S. "Improved nuclei segmentation by CNN-modified encoders in U-Net." NIBIB Building Bridges Across NIH and the Broader Engineering Community, Bethesda, Maryland, October 22, 2024. Abstracts. NIBIB Building Bridges Across NIH and the Broader Engineering Community, Bethesda, Maryland, October 22, 2024. , Oct-2024
Articles in Peer-reviewed Journals	Shrestha A, Bao X, Cheng Q, McRoy S. "CNN-modified encoders in U-Net for nuclei segmentation and quantification of fluorescent images." IEEE Access. 2024 Aug 1;12:107089-97. http://dx.doi.org/10.1109/ACCESS.2024.3436509 , Aug-2024